

Satellite Image Classification



EuroSAT Benchmark · CNN from Scratch · Transfer Learning

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27,000

Images

10

Classes

3

Models

96.26%

Peak Acc.

Project Structure

Four sections covering exploration, modelling, and evaluation

00



Exploratory Data Analysis

Dataset loading · Class distribution · Channel statistics · Data quality

01



Baseline CNN (Trained from Scratch)

Architecture design · Hyperparameter search · Learning curves · Evaluation

02



Transfer Learning

ResNet-50 (Microsoft) · ConvNeXT-tiny (Facebook) · Fine-tuning on EuroSAT

03



Model Comparison & Selection

Side-by-side metrics · Efficiency trade-offs · Final model choice

The EuroSAT Dataset

Sentinel-2 satellite imagery covering 10 land-use / land-cover classes

27,000

total RGB images

10

land-cover classes

64 × 64 px

image size (RGB)

0 – 255

pixel value range

70 / 30

train / val split

Seed 42

reproducible split

10 Land-Cover Classes

AnnualCrop

Forest

HerbaceousVegetation

Highway

Industrial

Pasture

PermanentCrop

Residential

River

SeaLake



Herbaceous-related
land plots

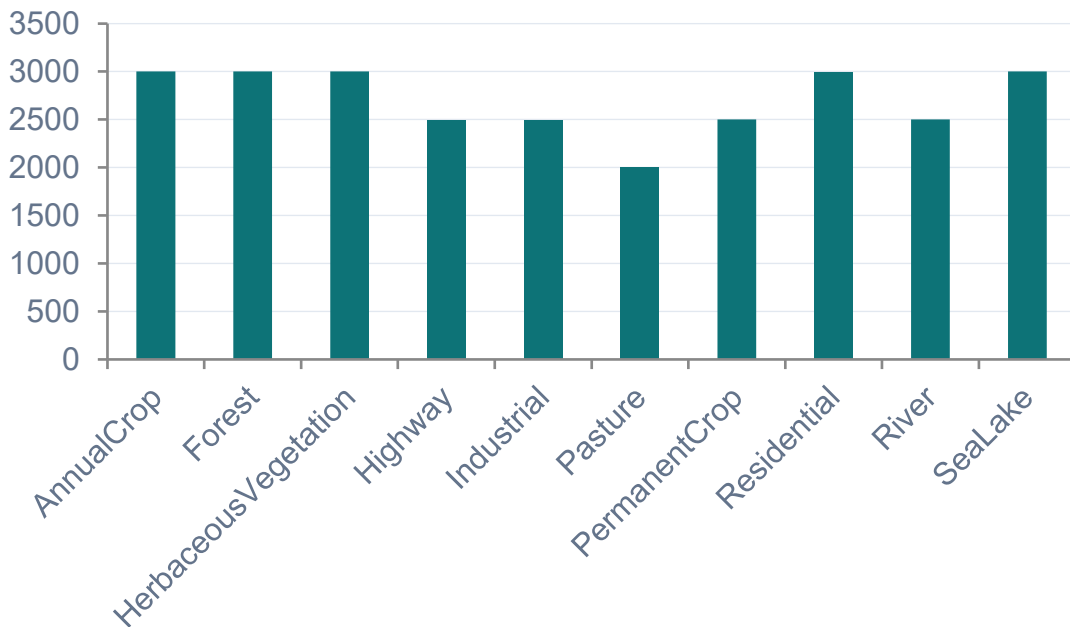


Non-Herbaceous-
related land plots

Class Distribution

EuroSAT is well-balanced — each class contributes approximately 10% of the dataset

Approx. Images per Class

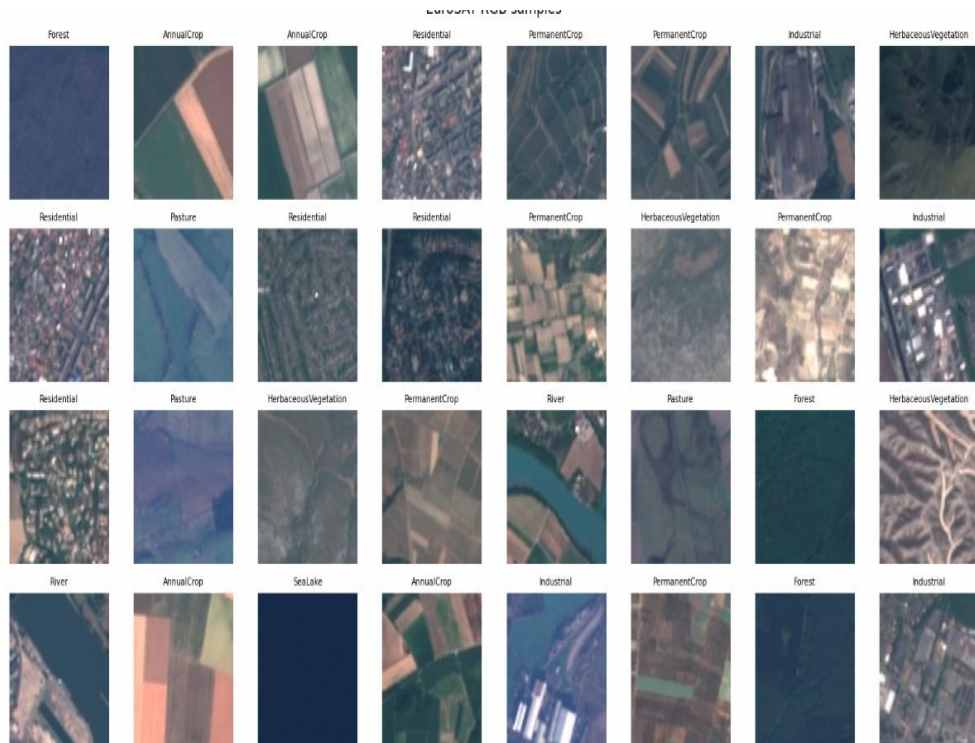


Key Findings

- ✓ All 10 classes are similarly represented (~10% each)
- ✓ Each class contains at least 2,000 images
- ✓ No class imbalance detected — no corrective action needed
- ✓ Overall accuracy is a reliable metric for this dataset
- ✓ No need for weighted loss or oversampling

Sample Images per Class

32 randomly sampled images across all 10 classes — EuroSAT RGB, 64×64 px



Key Observations

Visually distinct classes:

SeaLake (deep uniform blue), Industrial (grey grids), River (thin linear)

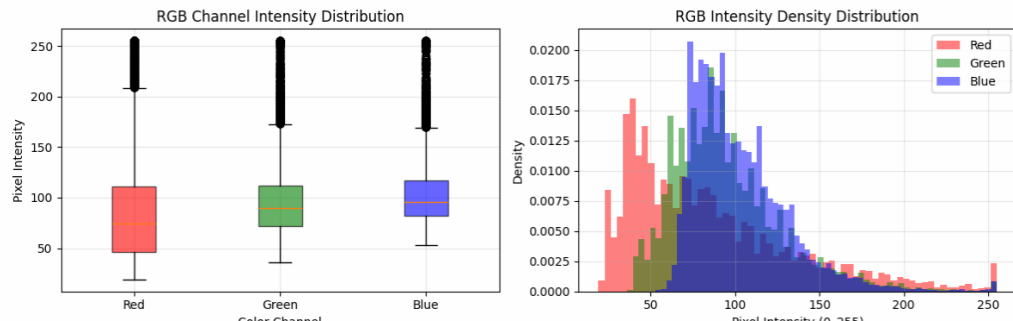
Ambiguous class pairs:

AnnualCrop vs PermanentCrop, Forest vs HerbaceousVegetation, Highway vs River

These pairs will be the main sources of confusion in the classifier — confirmed later by the confusion matrix.

Image Statistics & Normalisation

Per-channel mean and std computed over 2,048 training samples



Channel	Mean (0-255)	Std (0-255)	Mean (norm.)	Std (norm.)
Red	86.04	50.29	0.3374	0.1972
Green	95.75	34.04	0.3755	0.1335
Blue	103.00	28.79	0.4039	0.1129

Observations

- ✓ Channel means all sit in a similar range (86–103), indicating the dataset is moderately bright and balanced across RGB — no single channel dominates.
- ✓ Standard deviations are spread fairly evenly across channels (29–50), meaning variance is consistent.

Why It Matters

- ✓ These statistics are used to normalise inputs in all models, ensuring pixel values are centred around zero with unit variance.
- ✓ Stable gradient flow and faster convergence are direct benefits of this normalisation strategy.

Data Quality & Outlier Detection

Sanity checks confirmed a clean dataset before any modelling

0

Corrupt
(failed to load)

0

Wrong Shape
(not 3×64×64)

0

Blank Images
(all-black or white)

27,000

Total Images
Checked

Outlier Detection (Top 5% by Pixel Distance from Class Mean)

Metric	Value
Outliers detected (top 5%)	1,350 images
Distance threshold	3,370.48
Most variable classes	Highway, Residential
Root cause	Higher brightness due to varying light exposure
Action required	None — all images are labelled

Inter-class Visual Relationships

Inspecting sample and mean images highlights which class pairs are likely to challenge the classifier

Easy to Distinguish

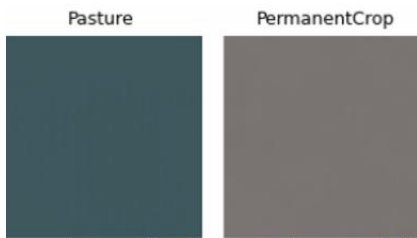
SeaLake — large uniform blue areas

River — thin elongated water signature

Industrial — grey rectangular structures



Visual Inspection



Mean Image

Harder to Separate

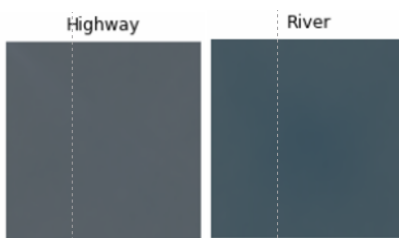
AnnualCrop vs PermanentCrop — both show geometric field patterns in muted greens and browns

Forest vs HerbaceousVegetation — dense green texture in both

Highway vs River — thin linear structures, visually similar at 64px resolution



Visual Inspection



Mean Image

Section 01

CNN

Training a compact CNN from scratch as a performance baseline



Baseline CNN

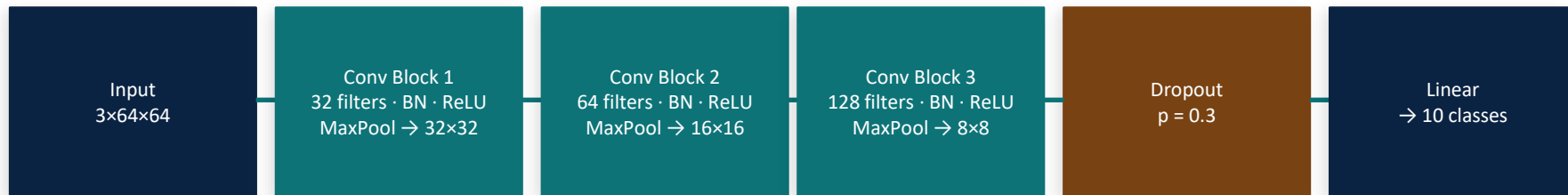
Custom architecture · 94,986 params

3 conv blocks trained from scratch
on EuroSAT only

Adam $lr=0.001$ · BatchNorm · Dropout $p=0.3$

CNN Architecture (Trained from Scratch)

A compact 3-block CNN with ~95K parameters designed as a robust baseline



Layer Summary

Layer	Type	Output Shape	Parameters
Conv1	Conv2d	(1, 32, 64, 64)	896
Bn1	BatchNorm2d	(1, 32, 64, 64)	64
Conv 2	Conv2d	(1, 64, 32, 32)	18,496
Bn2	BatchNorm2d	(1, 64, 32, 32)	128
Pool	MaxPool2d	(1, 32, 32, 32)	0
Conv3	Conv2d	(1, 128, 16, 16)	73,856
Conv3	Conv2d	(1, 128, 16, 16)	256
Pool	MaxPool2d	(1, 128, 8, 8)	0
Dropout	Dropout p=0.3	(1, 128)	0
Classifier	Linear	(1, 10)	1,290

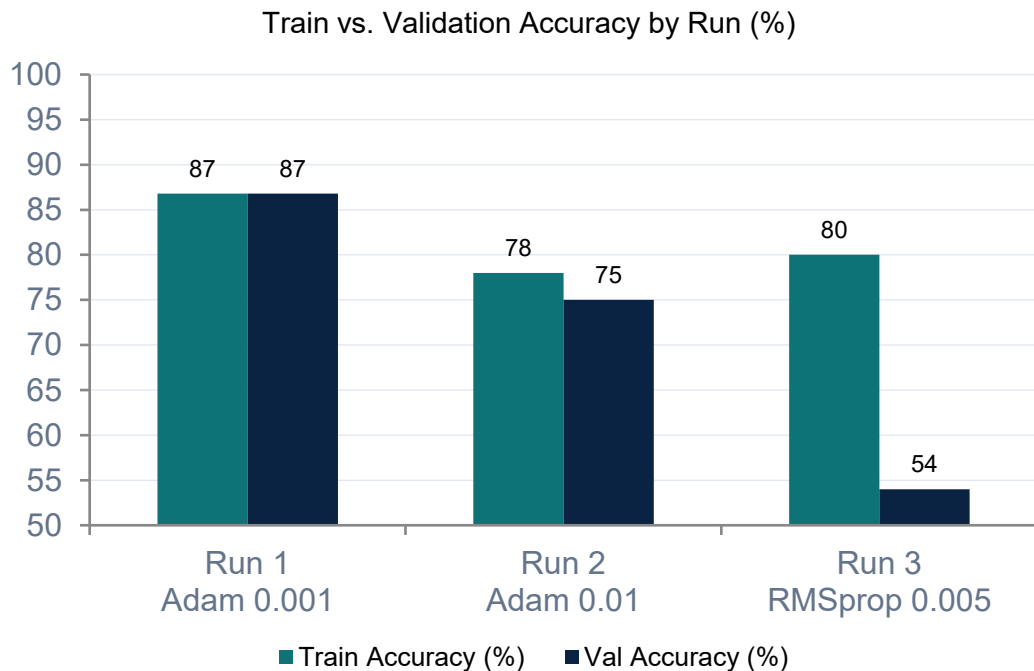
classifier → Linear (1, 10) | TOTAL TRAINABLE PARAMETERS: 94,986

Design Choices

- ✓ Filter doubling (32→64→128): learns increasingly abstract features
- ✓ Batch Normalisation: stabilises gradient flow
- ✓ Max Pooling: halves spatial size after each block
- ✓ Dropout (p=0.3): prevents overfitting before classifier

Hyperparameter Search Results

Three configurations evaluated — Run 1 selected as the best generalising model



#	Config	Epochs	Val Acc	Result
1	Adam / 0.001	12	87%	✓ Best
2	Adam / 0.01	4	76%	Oscillates
3	RMSprop / 0.005	7	54%	Overfit

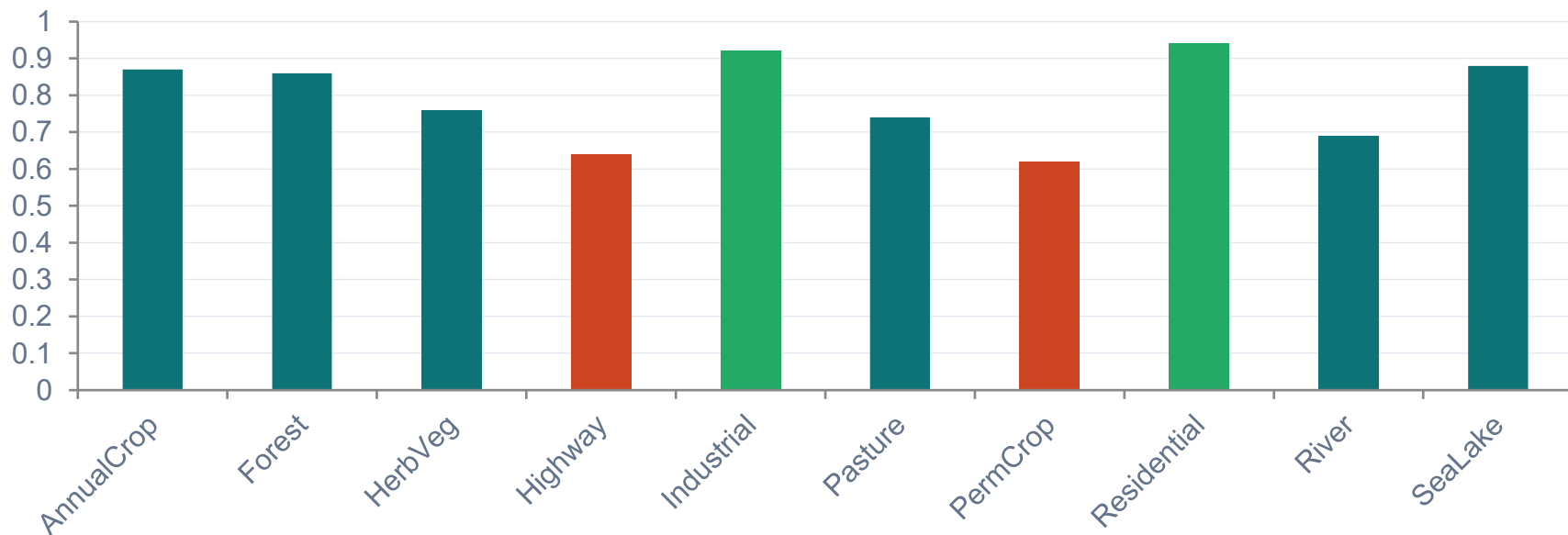
Takeaways

- ✓ Run 1 (Adam, lr=0.001) converges smoothly — ideal learning rate for this architecture
- ✓ Run 2 (lr=0.01) is too aggressive — oscillations in validation loss
- ✓ Run 3 (RMSprop) triggers early stopping after 3 epochs — fails to generalise (val acc 69%)

CNN Test Set Performance

Final evaluation on held-out validation set — Overall Accuracy: ~80%

F1-Score per Class (CNN Baseline)



Best Class: Industrial / Residential (F1 = ~0.95)

Hardest Class: Highway / PermCrop (F1 = ~0.65)

Section 02

Transfer Learning

Fine-tuning pretrained models on EuroSAT



ResNet-50

Microsoft · 23.5M params

Deep residual network
pretrained on ImageNet

Backbone frozen · Head replaced · Adam lr=0.001



ConvNeXT-tiny

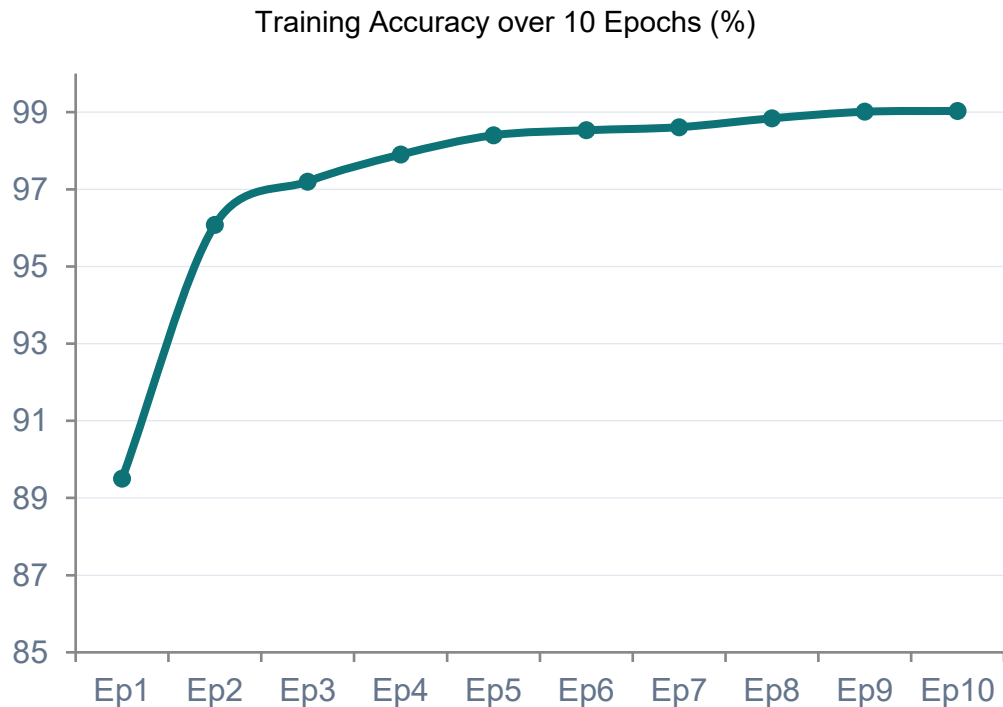
Facebook · 27.8M params

Modern CNN redesigned
through Vision Transformer lens

Backbone frozen · Head replaced · Adam lr=0.001

ResNet-50 — Pretrained Model A

microsoft/resnet-50 · ImageNet pretrained · Backbone frozen · 10 epochs · 23.5M trainable params)



Residual connections

Skip connections prevent vanishing gradients in deep 50-layer networks

Bottleneck blocks

3×3 conv sandwiched between 1×1 layers
reduces parameters while preserving depth

4 residual stages

Progressive feature maps:
64 → 128 → 256 → 512 channels

BatchNorm + ReLU

Stable training across all 50 layers

ConvNeXT-tiny — Pretrained Model B

facebook/convnext-tiny-224 · ImageNet pretrained · Head only (7,690 trainable params)

Training Accuracy over 5 Epochs (%)



7×7 kernels

Larger receptive field vs. 3×3 in ResNet

Depthwise conv

Fewer computations, more expressiveness

Layer Norm

Borrowed from Transformers — more stable

GELU activation

Smoother gradients than ReLU

Inverted bottleneck

Classification Reports — ResNet-50

96.26%

Test Accuracy

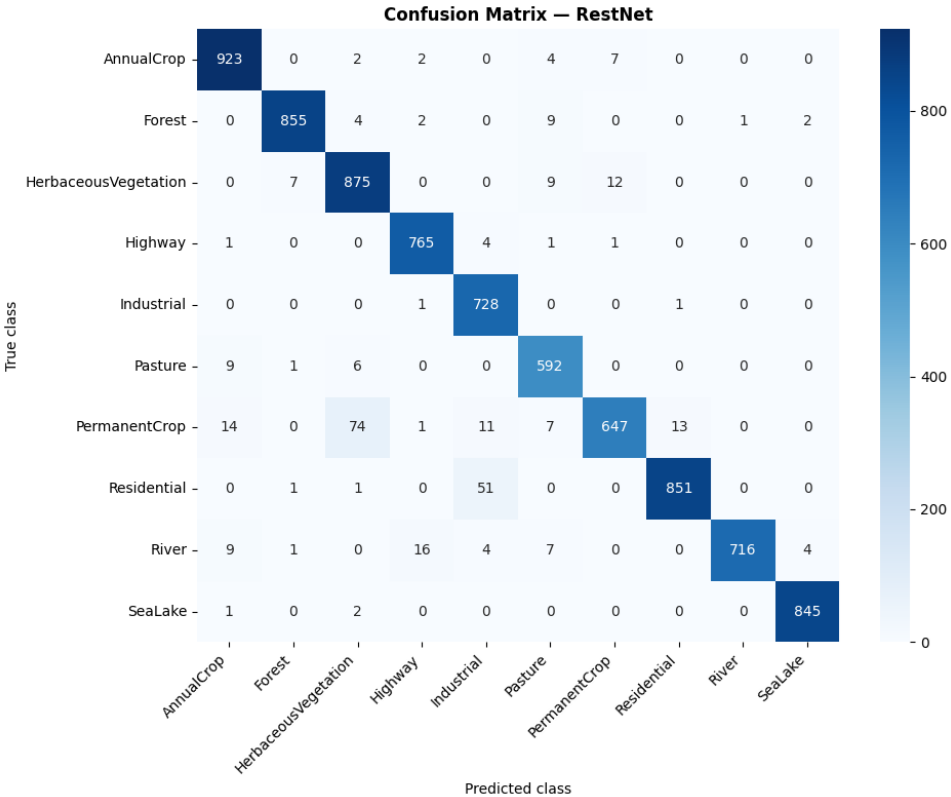
23.5M

Parameters

10

Epochs

Class	Precision	Recall	F1
AnnualCrop	96%	98%	97%
Forest	99%	98%	98%
HerbaceousVegetation	91%	97%	94%
Highway	97%	99%	98%
Industrial	91%	100%	95%
Pasture	94%	97%	96%
PermanentCrop	97%	84%	90%
Residential	98%	94%	96%
River	100%	95%	97%
SeaLake	99%	100%	99%



Classification Reports — ConvNeXT-tiny

95.16%

Test Accuracy

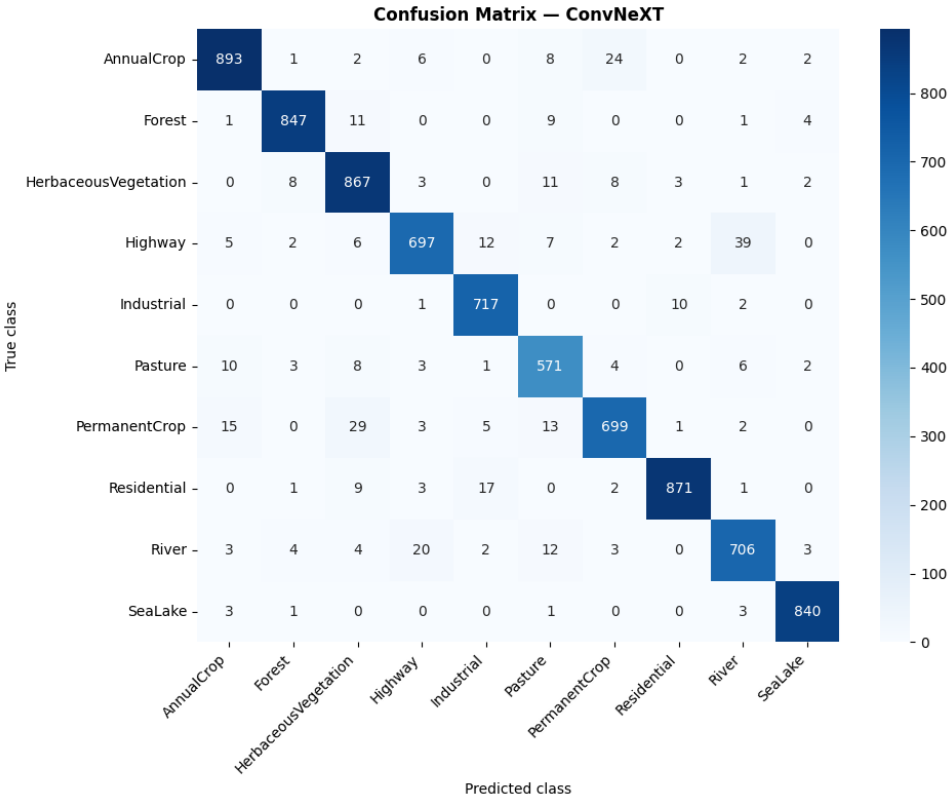
7,690

Trainable Params

5

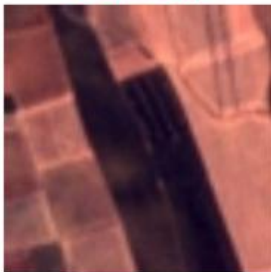
Epochs

Class	Precision	Recall	F1
AnnualCrop	96%	95%	96%
Forest	98%	97%	97%
HerbaceousVegetation	93%	96%	94%
Highway	95%	90%	92%
Industrial	95%	98%	97%
Pasture	90%	94%	92%
PermanentCrop	94%	91%	93%
Residential	98%	96%	97%
River	93%	93%	93%
SeaLake	98%	99%	99%

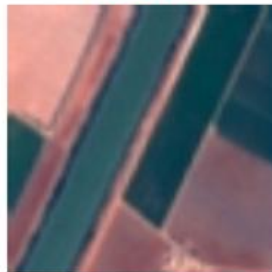


ConvNeXT-tiny - 10 most confident wrong predictions

Pred: PermanentCrop (99.5%)
True: AnnualCrop



Pred: AnnualCrop (98.8%)
True: River



Pred: PermanentCrop (98.6%)
True: AnnualCrop



Pred: Industrial (98.1%)
True: River



Pred: River (98.0%)
True: Highway



Pred: Pasture (97.8%)
True: Forest



Pred: HerbaceousVegetation (97.6%)
True: River



Pred: PermanentCrop (97.5%)
True: AnnualCrop



Pred: PermanentCrop (97.4%)
True: HerbaceousVegetation



Pred: River (97.3%)
True: Highway



Section 03

Model Comparison

Fine-tuning pretrained models on EuroSAT



Baseline CNN

Custom · 94,986 params

Custom · 94,986 params

*Backbone frozen · Head replaced · Adam
lr=0.001*



ResNet-50

Microsoft · 23.5M params

Deep residual network
pretrained on ImageNet

*Backbone frozen · Head replaced · Adam
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ConvNeXT-tiny

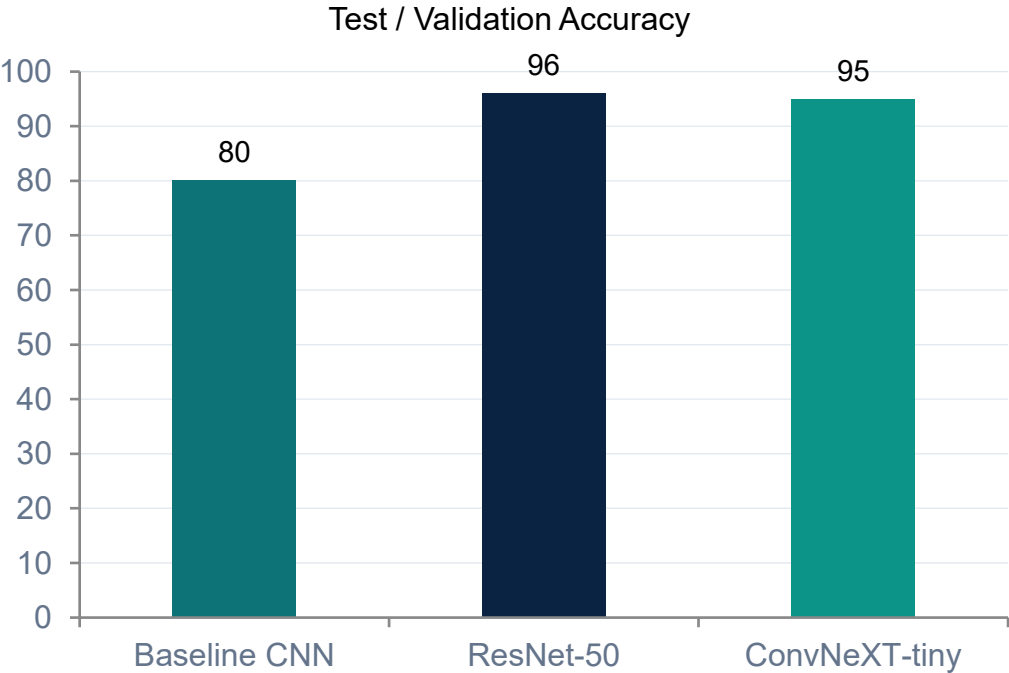
Facebook · 27.8M params

Modern CNN redesigned
through Vision Transformer lens

*Backbone frozen · Head replaced · Adam
lr=0.001*

Model Comparison

Performance, efficiency, and practical suitability across three models



Metric	Baseline CNN	ResNet-50	ConvNeX T-tiny
Test Accuracy	86.85%	96.26%	95.16%
Total Params	94,986	23.5M	27.8M
Trainable Params	94,986	23.5M	7,690
Epochs Required	12	10	5
Architecture	Custom CNN	Residual (2015)	Modern CNN (2022)
Pre-training Dataset	None	ImageNet	ImageNet

Conclusions



Dataset

EuroSAT is well-balanced and clean — all 27,000 images loaded correctly with no corrupted or blank files. The dataset is ready for modelling with no corrective action required.



Baseline CNN

The custom 3-block CNN achieves 80% accuracy on the validation set — a reasonable baseline for 94K parameters trained from scratch



Transfer Learning

ResNet-50 delivers 96.26% accuracy — a 29-point gain over the baseline. ConvNeXT-tiny achieves 95.16% with only 7,690 trainable parameters in 5 epochs, making it the most compute-efficient choice.



Recommendation

ConvNeXT-tiny is the recommended model: near-identical accuracy to ResNet-50 (95.16% vs 96.26%) with 3,000x fewer trainable parameters and half the training epochs. ResNet-50 is the alternative where maximum raw accuracy is the only criterion.

Next Steps & Recommendations

Unfreeze additional layers (progressive fine-tuning)

Both ResNet-50 and ConvNeXT-tiny were trained with the backbone fully frozen. Gradually unfreezing the last 2–3 blocks and fine-tuning with a lower learning rate (e.g. $1e-5$) could push accuracy above 97% without a significant computational cost increase

Address the hardest classes

Highway ($F1 = 0.92$ in ConvNeXT) and PermanentCrop ($F1 = 0.93$) are the weakest points across all models. Targeted augmentation for these classes (i.e., rotation, color jitter, random crops) could improve generalization on visually ambiguous samples

Expand the hyperparameter search

The current search was limited to 3 configurations. A broader grid (or random search) over batch size, weight decay, and dropout would give more confidence that the best configuration has been found.

Test on out-of-distribution data

EuroSAT covers a specific set of European regions. Testing the best model on satellite imagery from other geographies or seasons would reveal how well the learned features generalize beyond the training distribution.

Section 04

APP Build-up

Flask web app for real-time satellite image classification

